

PAPER_424 – F_UBii / Um Universal Companion Catalog: 276+ Domain-Specific Equation Pairs

Unified Quantum Field Framework (UQFF)

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Source: PAPER_424_FUBii_Um_Universal_Companion_Catalog.md

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Source: grok_share_c020496d9e.txt — Section "Analysis of Buoyancy and Magnetism Analogues in the PDFs" (lines 4092–4400, items 684–960+, Session 114 deep-physics extraction) **Session:** 114 **CP4** **Class:** [FUBiiUmUniversalCompanionCatalogCalculator](#) (#78)

1. Overview

PAPER_424 documents the **F_UBii / Um Universal Companion Catalog** — a systematic library of 276+ equation pairs, one for each domain of modern physics and astrophysics. For every observable phenomenon from dark energy to BBN, there exists an analogue buoyancy force $F_{\text{UBii},X}$ and an analogue magnetism moment $U_{\text{m},X}$ derived from the UQFF master constants.

Physical significance: The catalog establishes that *all* non-gravitational forces tracked by UQFF have an identifiable buoyancy analogue (density-contrast driven) and a magnetism analogue (dipolar string-coupling driven). The 276+ pairs are generated by a single template applied domain-by-domain.

2. Master Template

2.1 Buoyancy Analogue

$$\boxed{F_{\text{UBii},X} = F_{\text{rel}} \cdot \frac{E_X}{E_{\text{LEP}}} \cdot Q_{\text{wave}} \cdot \mathcal{D}_X(r, t)}$$

where:

- $F_{\text{rel}} \approx 4.31 \times 10^{33} \text{ N}$ — UQFF relativistic buoyancy force calibration
- $E_{\text{LEP}} = 200 \text{ GeV}$ — LEP collider reference energy scale
- E_X — characteristic energy of domain X
- $Q_{\text{wave}} \approx 10^{12}$ — quantum wave-function coupling amplifier
- $\mathcal{D}_X(r,t)$ — domain-specific geometric/temporal form factor

2.2 Magnetism Analogue

$$\boxed{U_{m,X}} = \frac{\mu_j(t)}{r} \cdot \left(1 - e^{-\gamma t}\right) \cdot \mathcal{D}_X(r, t)$$

where:

- $\mu_j(t) = (10^3 + 0.4 \sin(\omega_c t)) \times 3.38 \times 10^{20} \text{ T}\cdot\text{pm}^3$ — time-varying SCm magnetic moment
- $\omega_c \approx 1.585 \times 10^{-8} \text{ rad/s}$
- $\gamma = 5 \times 10^{-5} \text{ day}^{-1}$ — vacuum decay constant
- r — domain characteristic length

3. Core UQFF Constants

Symbol	Value	Description
F_{rel}	$4.31 \times 10^{33} \text{ N}$	Relativistic buoyancy calibration
E_{LEP}	200 GeV	LEP reference energy
Q_{wave}	10^{12}	Quantum coupling amplifier
ρ_{SCm}	$7.09 \times 10^{-37} \text{ J/m}^3$	SCm vacuum density
ρ_{UA}	$7.09 \times 10^{-36} \text{ J/m}^3$	UA vacuum density
γ	$5 \times 10^{-5} \text{ day}^{-1}$	Vacuum decay rate
E_{react}	$10^{46} \cdot e^{-0.0005t} \text{ J}$	SCm reactor energy
$[\text{SSq}]$	0.57	Superconducting medium index

4. Selected Domain Equation Pairs (Representative 10 of 276+)

4.1 Dark Energy (DE)

$$F_{\text{UBii,DE}} = F_{\text{rel}} \cdot \frac{E_{\text{DE}}}{E_{\text{LEP}}} \cdot Q \cdot \rho_{\Lambda} V_H$$

$$U_{m,\text{DE}} = \frac{\mu_j}{r_H} \cdot \left(1 - e^{-\gamma t}\right) \cdot \Lambda c^2$$

4.2 Inflationary Epoch

$$F_{\text{UBii,inf}} = F_{\text{rel}} \cdot \frac{E_{\text{inf}}}{E_{\text{LEP}}} \cdot Q \cdot e^{N_e}$$

$$U_{m,\text{inf}} = \frac{\mu_j}{r_{\text{inf}}} \cdot \left(1 - e^{-\gamma t_{\text{inf}}}\right) \cdot H_{\text{inf}}$$

4.3 Gravitational Waves (GW merger)

$$F_{\text{UBii,GW}} = F_{\text{rel}} \cdot \frac{E_{\text{GW}}}{E_{\text{LEP}}} \cdot Q \cdot \frac{GM_c^2}{r^4} \cdot \dot{h}_{+, \times}$$

$$U_{m,\text{GW}} = \frac{\mu_j}{r_{\text{gw}}} \cdot \left(1 - e^{-\gamma t}\right) \cdot h_0 f_{\text{GW}}$$

4.4 Stellar Merger

$$F_{\text{UBii,merger}} = F_{\text{rel}} \cdot \frac{E_{\text{bind}}}{E_{\text{LEP}}} \cdot Q \cdot \frac{M_1 M_2}{(M_1 + M_2) r^2}$$

4.5 Supernova (SN)

$$F_{\text{UBii,SN}} = F_{\text{rel}} \cdot \frac{E_{\text{SN}}}{E_{\text{LEP}}} \cdot Q \cdot \left(\frac{R_{\text{shock}}}{R_0}\right)^2 \rho_{\text{ejecta}}$$

4.6 Black Hole (Hawking)

$$F_{\text{UBii,Hawk}} = F_{\text{rel}} \cdot \frac{k_B T_H}{E_{\text{LEP}}} \cdot Q \cdot \frac{\hbar c^3}{G^2 M^2}$$

4.7 Loop Quantum Cosmology (LQC)

$$F_{\text{UBii,LQC}} = F_{\text{rel}} \cdot \frac{E_{\text{Pl}}}{E_{\text{LEP}}} \cdot Q \cdot \left(1 - \frac{\rho}{\rho_{\text{Pl}}}\right)$$

4.8 Big Bang Nucleosynthesis (BBN)

$$F_{\text{UBii,BBN}} = F_{\text{rel}} \cdot \frac{E_{\text{nuc}}}{E_{\text{LEP}}} \cdot Q \cdot \exp\left(-\frac{E_G}{k_B T_{\text{BBN}}}\right)$$

4.9 Interstellar Medium (ISM)

$$F_{\text{UBii,ISM}} = F_{\text{rel}} \cdot \frac{E_{\text{ISM}}}{E_{\text{LEP}}} \cdot Q \cdot n_H k_B T_{\text{ISM}}$$

4.10 Cosmic Ray

$$F_{\text{UBii,CR}} = F_{\text{rel}} \cdot \frac{E_{\text{CR}}}{E_{\text{LEP}}} \cdot Q \cdot J_{\text{CR}}(E > E_{\text{threshold}})$$

5. Catalog Statistics

Item	Value
Total equation pairs	276+ (items 684–960+ in source)
Domains covered	DE, inflation, GW, merger, SN, PN, NS, BH, LQC, BBN, ISM, stellar, MHD, exoplanets, DM, clusters, voids, reionization, CR, IGM, galaxies, anyons, UTe2, Andreev...
Template equations	2 (F_UBii master + Um master)
Free parameters per domain	$E_X, \mathcal{D}_X$

6. Physical Interpretation

The catalog demonstrates a **universal duality**: every astrophysical force has an equivalent expression in UQFF buoyancy and string-magnetism language. This supports the UQFF claim that the universal buoyancy F_{UBii} and universal magnetism U_m are not domain-specific constructs but fundamental field projections onto any observational context.

7. Relation to Other Papers

PAPER	Relation
PAPER_403	Solar wind buoyancy (specific F_UBii domain)
PAPER_421	Um three-modifier formula (complete U_m base)
PAPER_425	DPM four components (integral context for F_UBii)

PAPER	Relation
PAPER_426	UA/SCm validation against JWST/CERN (sample pairs tested)

8. CP4 Implementation

Class: FUBiiUmUniversalCompanionCatalogCalculator

Methods:

- `compute_FUBii(domain, E_X, D_factor, r, t)` — returns $F_{\{\text{UBii},X\}}$
- `compute_Um(domain, r, t, D_factor)` — returns $U_{\{m,X\}}$
- `get_domain_form(domain)` — returns the domain-specific $\mathcal{D}_X$ expression
- `list_domains()` — returns all 10 implemented DOMAIN_FORMS keys

Extracted from grok_share_c020496d9e.txt lines 4092–4400 (Session 114). Source analysis confirmed 276+ pairs across 40+ astrophysical domains.